

1. Find the general antiderivative of the following functions.

1). $f(x) = \frac{1}{2} + \frac{3}{4}x^2 - \frac{4}{5}x^3$

$$F(x) = \frac{1}{2}x + \frac{3}{4} \cdot \frac{1}{3}x^3 - \frac{4}{5} \cdot \frac{1}{4}x^4 + C$$

$$= \frac{1}{2}x + \frac{1}{4}x^3 - \frac{1}{5}x^4 + C$$

2). $f(x) = 6\sqrt{x} - \sqrt[6]{x}$

$$= 6x^{1/2} - x^{1/6}$$

$$F(x) = 6 \left(\frac{2}{3}\right)x^{3/2} - \frac{6}{7}x^{7/6} + C$$

$$= 4x^{3/2} - \frac{6}{7}x^{7/6} + C$$

3). $f(x) = \sqrt[4]{x^3} + \sqrt[3]{x^4}$

$$= x^{3/4} + x^{4/3} + x^{1/2} + 3x^{1/2}$$

$$F(x) = \frac{4}{3}x^{7/4} + \frac{3}{4}x^{7/3} + 2x^{3/2} + 3 \cdot \frac{2}{3}x^{3/2} + C$$

$$= \frac{4}{3}x^{7/4} + \frac{3}{4}x^{7/3} + 2\sqrt{x} + 2x^{3/2} + C$$

4). $f(x) = \frac{2}{\sqrt{1-x^2}}$

$$F(x) = 2 \arcsin(x) + C$$

2. Find the following indefinite integrals.

1). $\int (3e^x + 7\sec^2 x) dx$

$$= 3e^x + 7\tan x + C$$

2). $\int (\cos \theta - 5 \sin \theta) d\theta$

$$= \sin \theta + 5 \cos \theta + C$$

$$3). \int \frac{5 - 4x^3 + 2x^6}{x^6} dx$$

$$= \int \left(\frac{5}{x^6} - \frac{4}{x^3} + \frac{2}{x} \right) dx$$

$$= \int \left(5x^{-6} - 4x^{-3} + \frac{2}{x} \right) dx$$

$$= 5 \cdot \left(-\frac{1}{5}\right) x^{-5} - 4 \cdot \left(-\frac{1}{2}\right) x^{-2} + 2 \ln|x| + C$$

$$= -\frac{1}{x^5} + 2 \cdot \frac{1}{x^2} + 2 \ln|x| + C$$

$$4). \int \frac{2 + x^2}{1 + x^2} dx$$

$$= \int \frac{1 + (1 + x^2)}{1 + x^2} dx = \int \frac{1}{1 + x^2} dx + \int \frac{1 + x^2}{1 + x^2} dx$$

$$= \arctan x + \int 1 dx$$

$$= \arctan x + x + C$$

3. Given $f''(x) = 4 \sin x + 3 \cos x$, $f(0) = 3$, $f'(0) = 4$, find $f(x)$.

$$f'(x) = \int (4 \sin x + 3 \cos x) dx = -4 \cos x + 3 \sin x + C$$

$$f'(0) = -4 \cos 0 + 3 \sin 0 + C = 4 \Rightarrow -4 + C = 4 \Rightarrow C = 8$$

$$\Rightarrow f'(x) = -4 \cos x + 3 \sin x + 8$$

$$f(x) = \int (-4 \cos x + 3 \sin x + 8) dx = -4 \sin x - 3 \cos x + 8x + D$$

$$f(0) = -4 \sin 0 - 3 \cos 0 + 0 + D = -3 + D = 3 \Rightarrow D = 6$$

$$\Rightarrow f(x) = -4 \sin x - 3 \cos x + 8x + 6$$